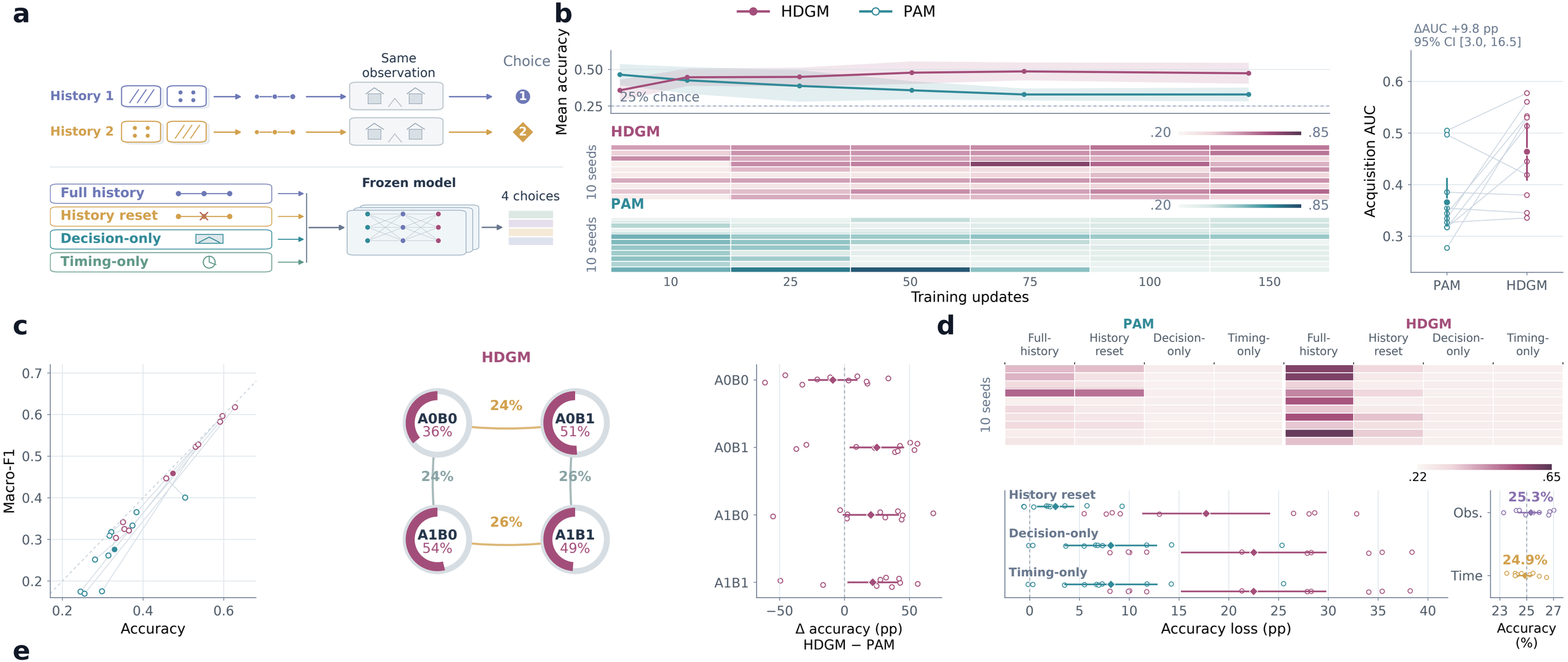

a
History 1
History 2
Same observation
Choice
Full history
History reset
Decision-only
Timing-only
Frozen model
4 choices
b
HDGM
PAM
Mean accuracy
0.50
0.25
25% chance
HDGM
PAM
.20
.85
10 seeds
10
25
50
75
100
150
Training updates
ΔAUC +9.8 pp
95% CI [3.0, 16.5]
Acquisition AUC
0.6
0.5
0.4
0.3
PAM
HDGM
c
Macro-F1
0.7
0.6
0.5
0.4
0.3
0.2
Accuracy
HDGM
A0B0
36%
A0B1
51%
A1B0
54%
A1B1
49%
24%
24%
26%
26%
A0B0
A0B1
A1B0
A1B1
−50
0
50
Δ accuracy (pp)
HDGM − PAM
d
PAM
HDGM
Full-history
History reset
Decision-only
Timing-only
10 seeds
.22
.65
History reset
Decision-only
Timing-only
0
5
10
15
20
25
30
35
40
Accuracy loss (pp)
25.3%
Obs.
24.9%
Time
23
25
27
Accuracy (%)
e
Cue A
Delay 1
+ k matched repeats
Cue B
Delay 2
+ k matched repeats
Decision
0
4
8
16
32
64
50%
25%
k repeats / delay
Schematic
k
HDGM
PAM
0 47.3% 32.7%
4 47.4% 26.8%
8 47.5% 26.6%
16 47.4% 26.6%
32 47.5% 26.6%
64 47.4% 26.6%
Δ accuracy (pp)
k = 64 − 0
5
0
−5
−15
−25
HDGM
PAM
ΔΔ = +6.1 pp
p = .023; Holm p = .070